

ChatGPT 5.1 ▾

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I am creating an AI-enhanced learning tool with ChatGPT "project" workplace for my deep learning class, and I want the structure of the guided conversation to follow the way of thinking reflected in the course materials — including the lecture notes, discussion notes, and homework solutions I will attach.

These materials have a specific pattern:

- start with intuition,
- then define notation clearly,
- then expand formulas step-by-step,
- then analyze complexity using consistent assumptions,
- and finally compare different mechanisms

I want to design a reusable sequence of prompts or questions that my classmates can follow to learn the concepts of:

1. Different attention mechanisms (MHA, MQA, GQA, causal, cross-attention in lecture and discussion)
2. Computational complexity of attention (like in the provided question set)
3. Kernelized / linear attention (as in the homework section)

****My question:****

> With ChatGPT "project" workplace, how should I design and order my prompts so that the model walks through the material using the same reasoning structure

as the course, and respects the way the instructors expect us to think through problems, as seen in the attached source questions?

>

****Please propose:****

- an ordered sequence of prompts to be used in this project,
- and suggestions for how to ensure the model avoids hallucinations and sticks to the course's notation and assumptions.

Below is a **ready-to-use, ordered prompt sequence** you can place inside a ChatGPT "project" workspace so that every guided conversation follows the *exact reasoning structure* expected in your deep learning class (intuition → notation → derivation → complexity → comparison).

I also provide **meta-prompts** that constrain hallucination, enforce reliance on your uploaded course materials, and ensure the model always uses the instructors' notation/assumptions.

Part 1 — Overall Structure of Your Project Workspace

In ChatGPT **Project**, you can create a "Guided Workflow" using a sequence of reusable prompts.

Your sequence should mirror the structure of your course materials:

1. Intuition
2. Notation
3. Step-by-step derivation
4. Complexity analysis

5. Comparison to alternatives
6. (Optional) Problem-solving walkthrough in the style of homework/discussion

Your project should therefore define a *fixed template the model must follow* for all attention-related topics.

✅ Part 2 — Ordered Sequence of Prompts for Attention Mechanisms

Below is a **fully designed prompt chain** you can paste into your Project as sequential blocks.

Each block includes the *exact kind of student question* and the *meta-instruction* to ChatGPT so it follows the class reasoning.

Prompt 1 — “Follow-the-course-thinking” Instruction (Global Rule)

Place this as a system prompt at the top of the Project.

SYSTEM RULE:

When teaching or solving any problem, strictly follow the reasoning structure used in the uploaded course materials (lecture notes, discussions, homework solutions):

1. “Start with intuition”
2. “Then define notation precisely, following the same symbols and conventions as the course”
3. “Then derive formulas step-by-step”
4. “Then analyze computational complexity using the same assumptions as the course”
5. “Then compare with related mechanisms”

If a user asks outside this structure, *adapt the answer into the course’s structure*. Cite definitions or assumptions explicitly from the uploaded materials instead of hallucinating.

If something is missing in the materials, say so and propose the closest valid

interpretation.

Section A — Prompts for Learning Different Attention Mechanisms

Prompt 2 — Intuition first

"Give the intuition behind [MHA / MQA / GQA / cross-attention / causal attention], following the intuition style used in the course discussion notes. Do NOT define formulas yet. Only focus on the high-level idea and what problem the mechanism is solving."

Prompt 3 — Notation Setup

***"Now set up the notation *exactly* as used in the lecture notes/discussion/hw solutions:

- "define Q, K, V "
- "define dimensions d, d_k, d_v "
- "define number of heads h (if applicable)"
- "define merge/reshape conventions"
- "define assumptions such as sequence length n "

Be consistent with the course notation; do NOT invent new symbols."**

Prompt 4 — Step-by-step Derivation

***"Now derive the formulas step-by-step, following the style of the course:

- "start from Q, K, V "
- "express attention scores"

- "show softmax"
- "expand multi-head (or multi-query / group-query) structure"
- "show how projections differ"
- "show output combination"

Write every step explicitly as in homework solutions."**

Prompt 5 — Computational Complexity

***"Give the computational complexity of this attention mechanism using the *same assumptions* as the course's complexity analysis:

- "sequence length n "
- "hidden size d "
- "head dimension d_k "

Follow exactly the style of the provided computational complexity question set."**

Prompt 6 — Comparison Prompt

***"Compare this mechanism with the baseline (usually MHA) across:

- "parameter count"
- "computational cost"
- "memory cost"
- "qualitative trade-offs"

Use the comparison tables and wording style seen in lecture/discussion."**

Section B — Prompts for Kernelized / Linear Attention

Prompt 7 — Kernelized Attention Intuition

***"Explain the kernelized/linear attention intuition in the same high-level style used in homework sections:

- "motivation (handling $O(n^2)$ cost)"
- "feature maps ϕ "
- "associative rearrangement

Avoid formulas here."***"

Prompt 8 — Notation & Derivation for Linear Attention

***"Now reproduce the notation and step-by-step derivation of linear attention:

1. "define $\phi(\cdot)$ feature map"
2. "rewrite attention as $\Phi(Q) \cdot \Phi(K)^T$ "
3. "show low-rank rearrangement"
4. "derive $O(n)$ computation

Follow the step structure used in homework solution format."***"

Prompt 9 — Complexity Comparison

***"Compare standard softmax attention with linear/kernelized attention on:

- "time complexity"
- "memory complexity"
- "when linear attention is advantageous"

Use the same assumptions as the course and produce a short table."***"

Section C — Prompts for Ensuring Course-Style Thinking in Problem Solving

These prompts replicate the "reasoning steps" seen in the HW + discussion solutions.

Prompt 10 — Problem-Solving Template

******"Solve the following problem by following the structure used in the course solutions:

1. "Intuition"
2. "Notation"
3. "Step-by-step manipulation"
4. "Complexity assessment"
5. "Summary of key idea"

Problem: [insert hw-style question]"**

Section D — Anti-Hallucination Controls inside the Project

Include the following as **persistent meta-instructions** inside the project workspace.

Prompt 11 — "Use Source Material Only" Meta-rule

"When answering, rely strictly on the notation, formulas, and assumptions shown in the uploaded lecture notes, discussions, and homework solutions.

If something is not provided in these materials, say 'The course materials do not specify this; the closest standard interpretation is...'

Never guess or introduce new conventions."

Prompt 12 — "Check for Consistency" Meta-rule

******"Before giving the final answer, verify that:

1. "All notation matches the course"
2. "Each derivation step is valid"
3. "Complexity analysis uses identical assumptions as the course"
4. "No extra mechanisms or symbols not present in the course are introduced."

Section E — Optional Workflow Enhancements

Prompt 13 — Self-correction Layer

******Review your previous answer and identify:

- "inconsistencies with course notation"
- "steps that should be broken down more"
- "assumptions that should be stated explicitly"

Revise the answer following the style of course solution."******

Prompt 14 — Mini-Exam Mode

"Ask me 5 questions in the same style as our course HW or discussion section on this topic. After I answer, grade using the course rubric (show intermediate steps)."



Part 3 — Summary of the Complete Prompt Chain

Stage	Purpose	Prompt
Global meta-rule	Enforce course reasoning	Prompt 1
Intuition	Start like lecture/discussion	Prompt 2
Notation	Make symbols consistent	Prompt 3
Derivation	Follow hw steps	Prompt 4
Complexity	Use consistent assumptions	Prompt 5
Comparison	Compare mechanisms	Prompt 6
Kernelized intuition	Match hw	Prompt 7
Kernelized derivation	Step-by-step	Prompt 8
Kernelized complexity	Match hw	Prompt 9
Problem-solving template	Reproduce hw style	Prompt 10
Anti-hallucination	Enforce correctness	Prompts 11–12
Self-correction	Improve answers	Prompt 13
Exam mode	Practice	Prompt 14

You can convert these into “Project nodes” in ChatGPT and reuse them for every attention-related topic.

If you'd like

I can also: